

ARISTOTLE UNIVERSITY OF THESSALONIKI
FACULTY OF SCIENCES
DEPARTMENT OF MATHEMATICS

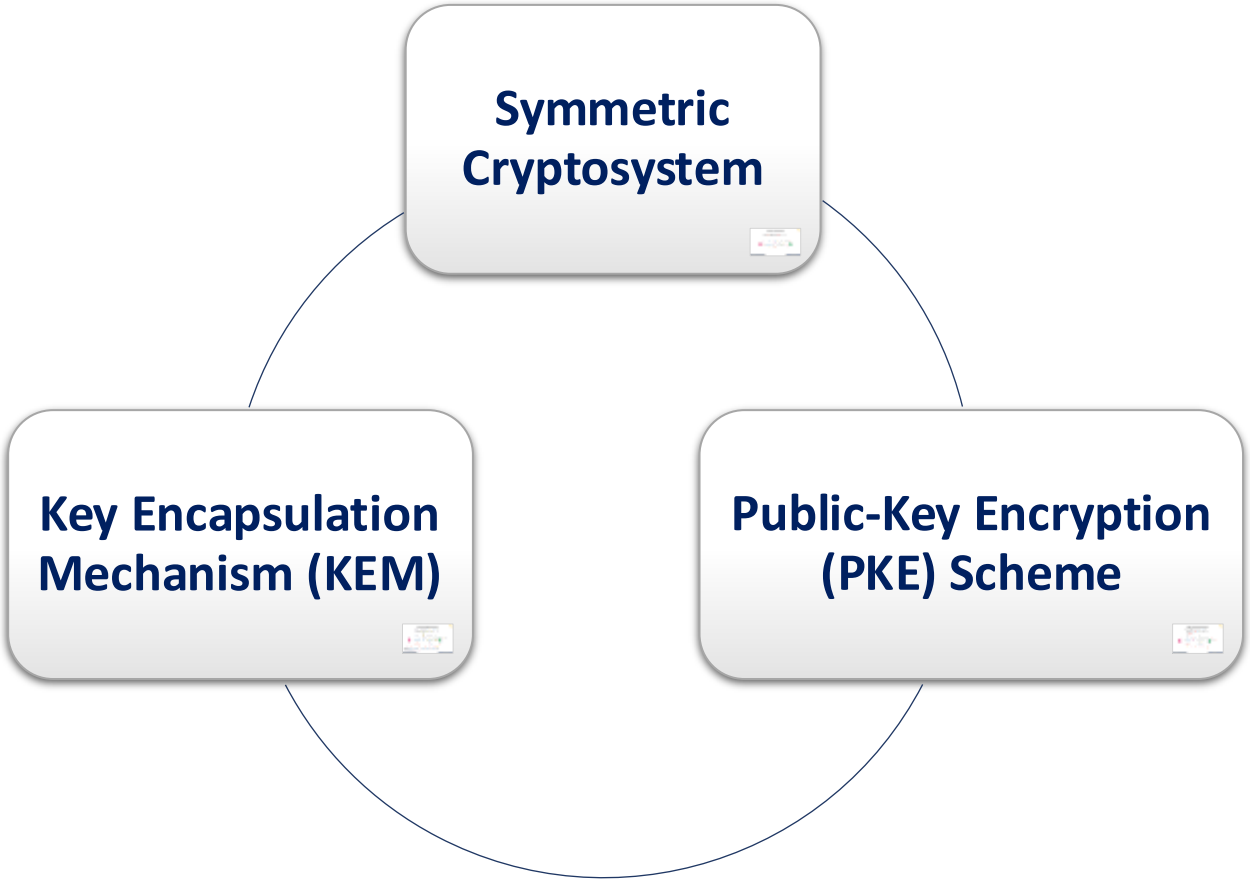


# Cryptography Based on Learning With Errors

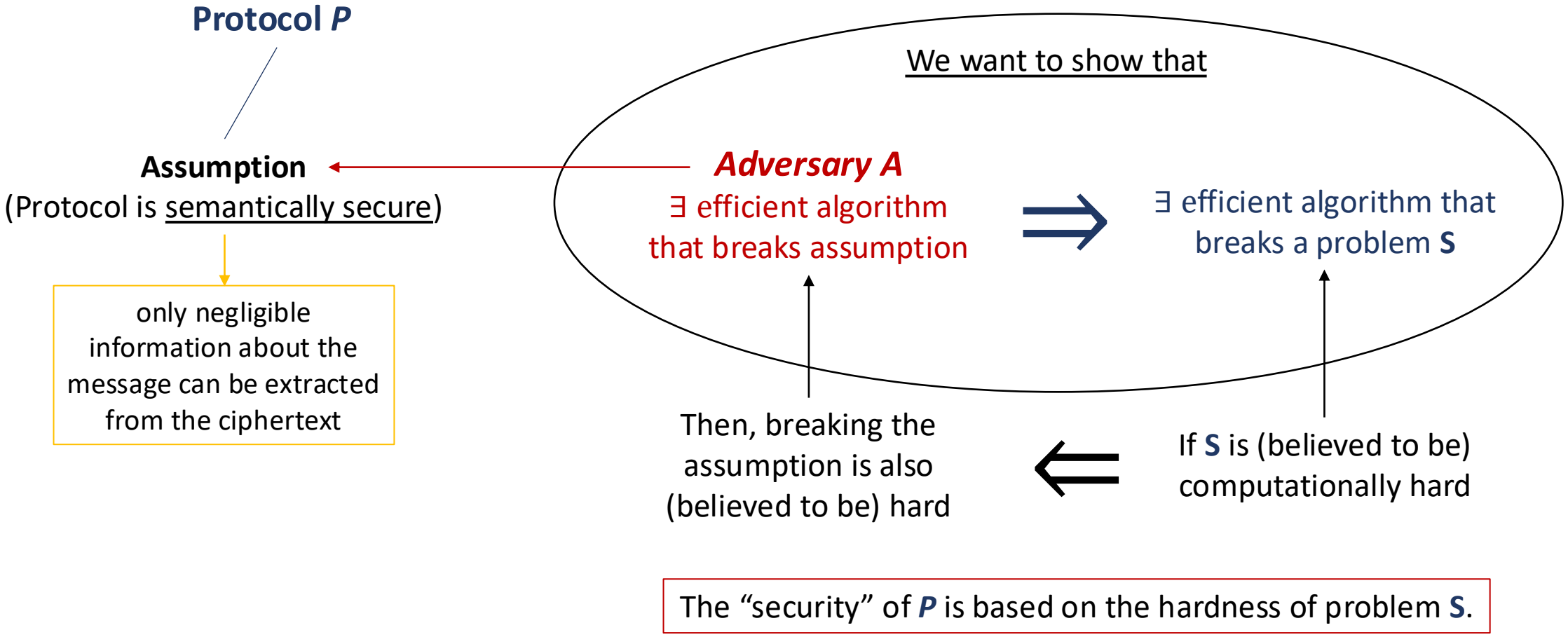
Florias Papadopoulos

Thessaloniki, 21 December 2023

# Cryptographic Schemes



# Provable Security of Cryptographic Schemes




The "security" of ***P*** is based on the hardness of problem **S**.

# Computationally Hard Problems: Today and Beyond

| | | |
|---|---|---|
| Problems used in Modern Cryptography | Integer Factorization | RSA, Rabin Cryptosystem |
| | Discrete Logarithm Problem (DLP) | Diffie-Hellman Key Exchange |
| | Elliptic Curve Discrete Logarithm Problem (ECDLP) | Elliptic-curve Diffie–Hellman (ECDH) |



| | | |
|---|---|---|
| Problems used in Post-Quantum Cryptography | Hard Problems in Lattices (SIS, LWE -> SVP, CVP) | NTRU, Kyber, Dilithium |
| | Finding solutions of multivariate equations | Rainbow |
| | Problem in isogenies of graphs of elliptic curves on finite fields | CSIDH, SIDH, SIKE |
| | Hardness of Decoding random linear codes | McElliece Cryptosystem |
| | Security of an underlying hash function | SPHINCS+ |

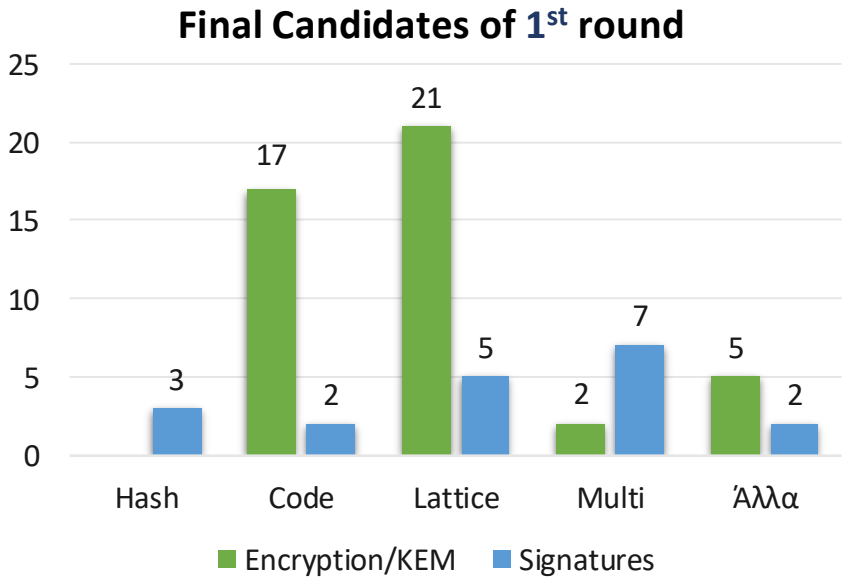
# NIST's Post-Quantum Competition

- **National Institute of Standards and Technology (NIST)**

**NIST's Competition (2016)**

New post-quantum standards for:

- Public-Key Encryption (PKE)
- Key Encapsulation Mechanisms (KEM)
- Digital Signatures

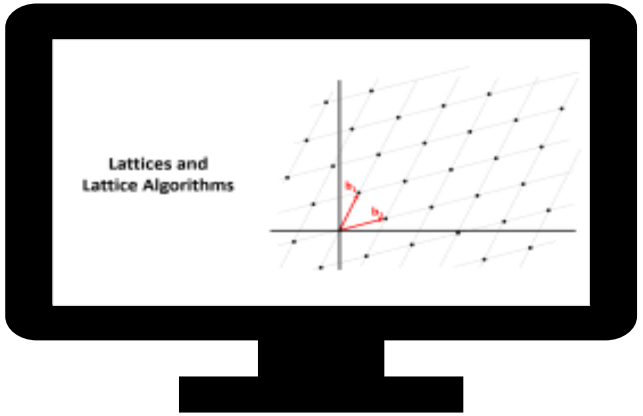



(**2023**) Finalist for general encryption: **Kyber**

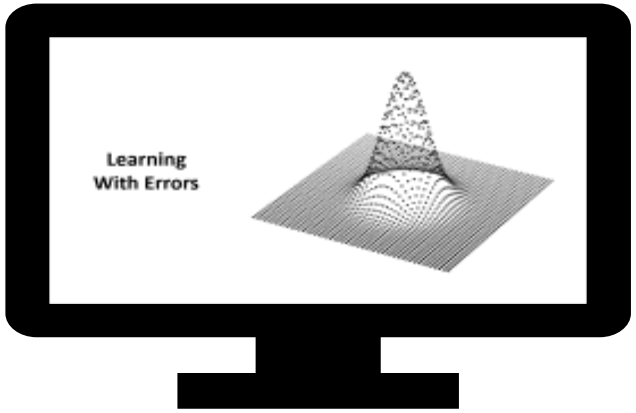
(**2024**) Standardization

KEM based on Module-LWE

Learning With Errors in module lattices




Lattices



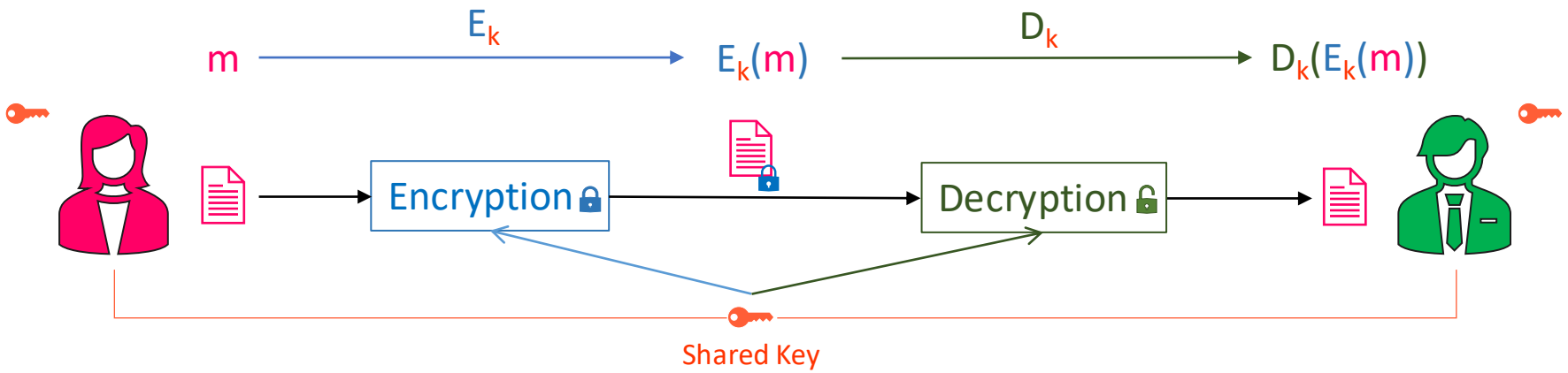

Learning With Errors




CRYSTALS-Kyber

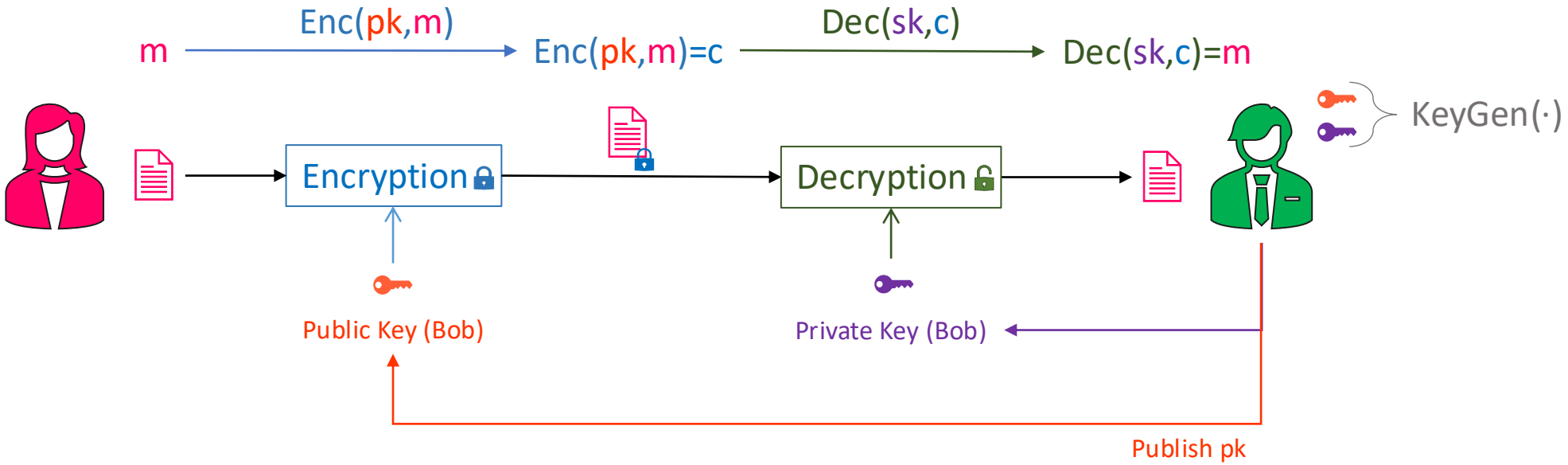
# Symmetric Cryptosystem

- One shared key is used for both encryption and decryption.
- The shared key is agreed upon beforehand.



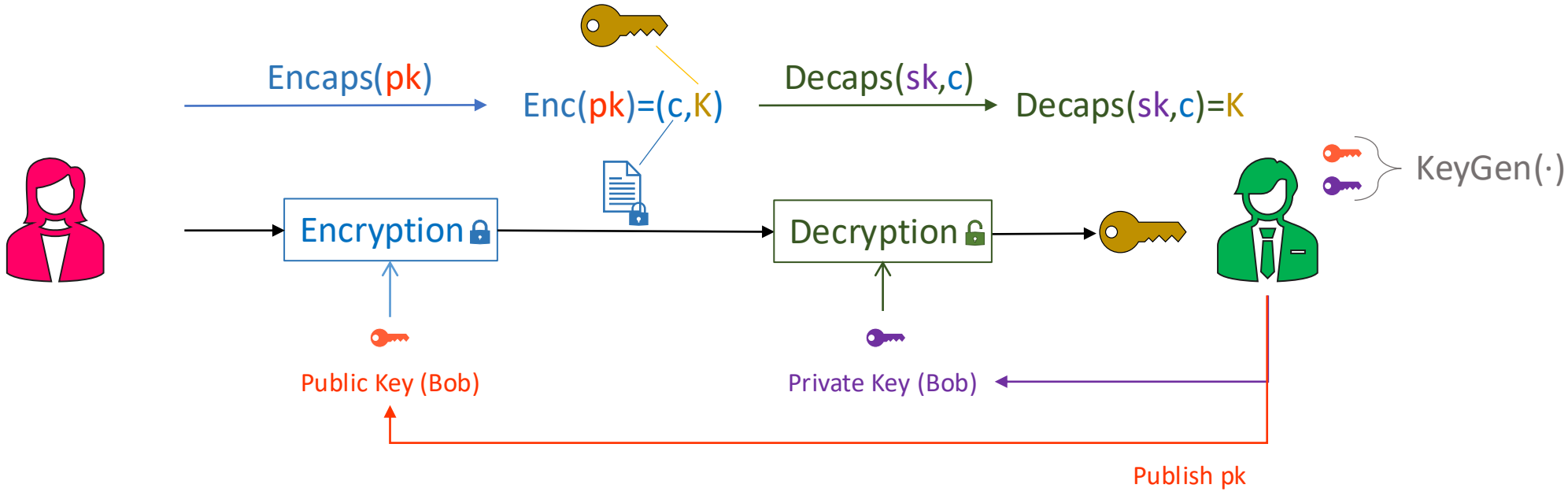
# Public-Key Encryption Scheme

- PKE consist of three algorithms: KeyGen(·), Enc(·), Dec(·)
- Two different keys are used for encryption and decryption:
  - A **private key**
  - A **public key**



# Key Encapsulation Mechanism

- KEM consist of three algorithms: KeyGen(·), Encaps(·), Decaps(·)
- A KEM is used to establish a shared secret key.




Not safe from
- Man-in-the-Middle attacks (MitM)
- Impersonation attacks

→ Authenticated Key Exchange (AKE) protocols

# What is the state of quantum computing?



TOM SIMONITE BUSINESS OCT 24, 2019 6:52 PM

**Quantum Computing Is Here! But Also Not Really**

Google shows off its Sycamore chip that it says reached "quantum supremacy," but it will still be years before quantum computing is useful.

Wired

QUANTUM COMPUTING | OPINION

**Will Quantum Computing Ever Live Up to Its Hype?**

One expert warns that the field is overpromising, while another says his firm is on the verge of building "useful" machines

By John Horgan on April 20, 2021

Scientific American

AI

**The Quantum Era Is Coming, But It's Not Here Yet**

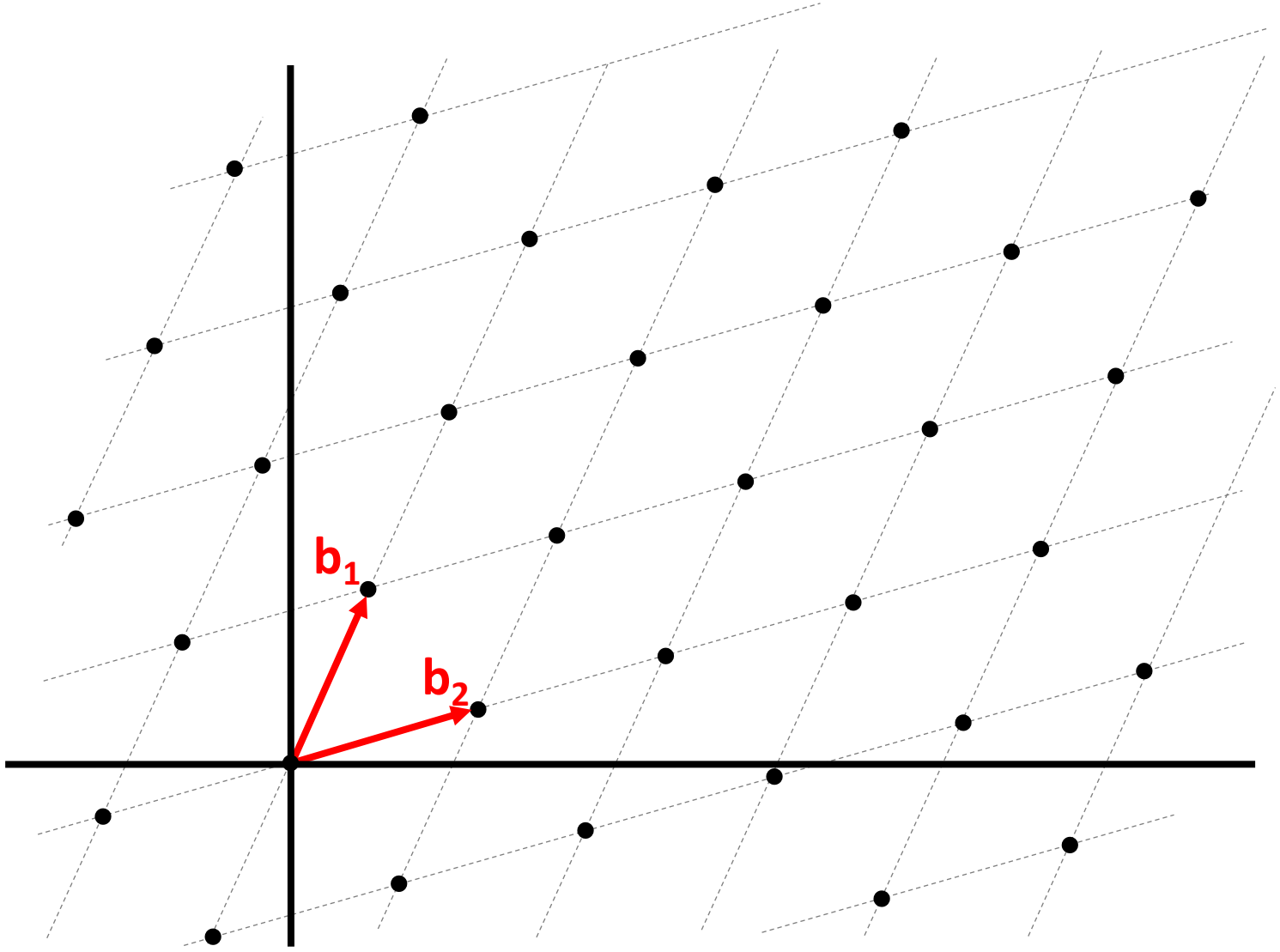
Forbes

Why care ?

↓

**STORE NOW, DECRYPT LATER**

# Lattices and Lattice Algorithms



# Basics of Lattices II

lattice rank

A subset $L \subset \mathbb{R}^m$ is called a **lattice**, if there is a linearly independent set of vectors $\{\boldsymbol{b}_1, \dots, \boldsymbol{b}_n\}$ in $\mathbb{R}^m$ $(m \geq n)$ s.t.:

$$L = L(\boldsymbol{b}_1, \dots, \boldsymbol{b}_n) = \left\{ \sum_{j=1}^{n} x_j \boldsymbol{b}_j \mid x_j \in \mathbb{Z} \right\}$$

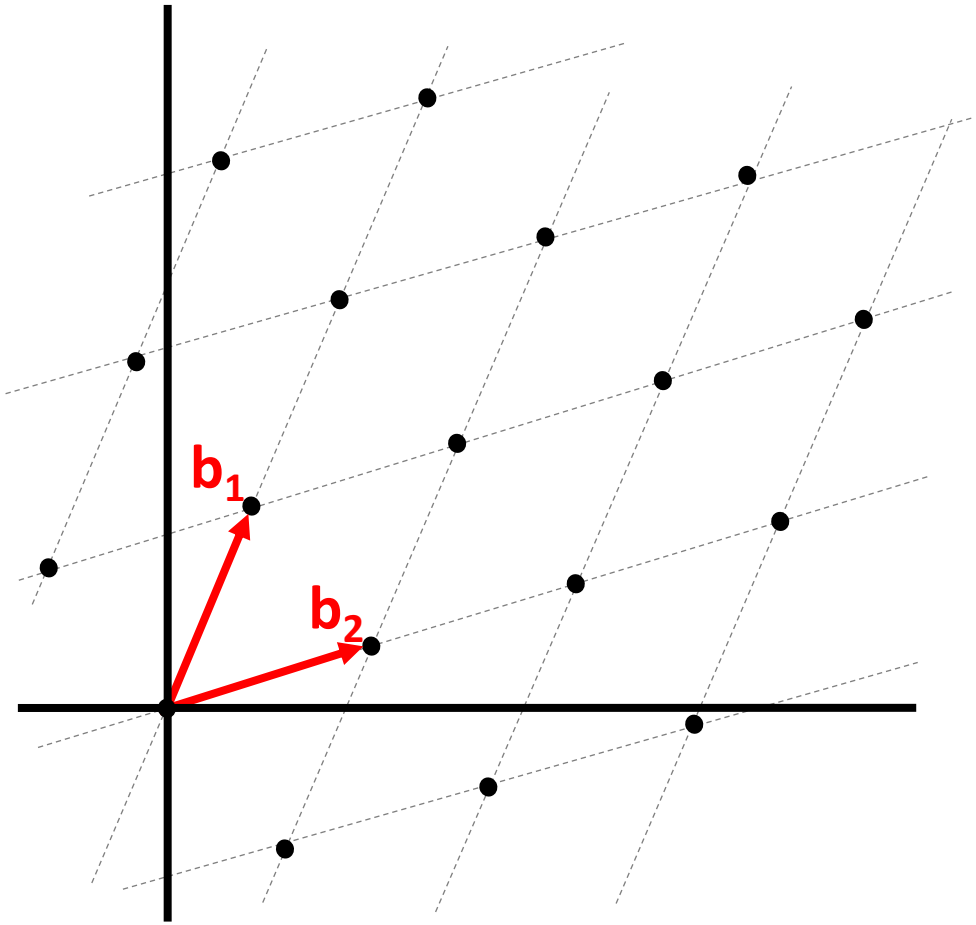
lattice basis

lattice dimension

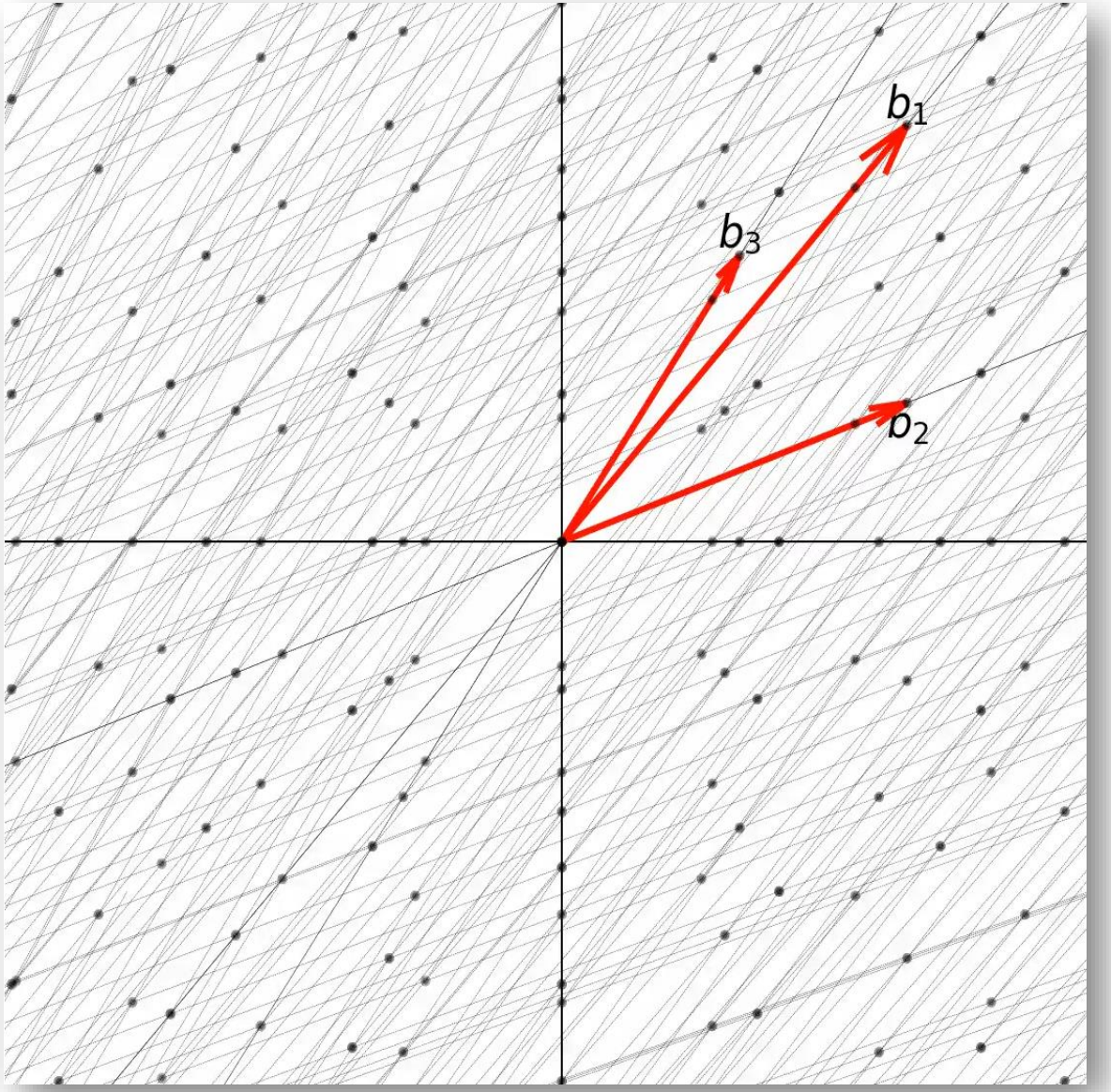
$$L(\boldsymbol{B}) = \{\boldsymbol{Bx} \mid \boldsymbol{x} \in \mathbb{Z}^n\} = \boldsymbol{B}\mathbb{Z}^n$$

- If $n = \mathrm{m} \rightarrow L$ is a **full-rank lattice**.
- If $n > 2 \rightarrow$ infinitely many choices for a basis.
- $\boldsymbol{B} \leftrightarrow \boldsymbol{B}'$ easily.

$$L = \{x_1\boldsymbol{b}_1 + x_2\boldsymbol{b}_2 \mid x_1, x_2 \in \mathbb{Z}\}$$



# An example of a 3D lattice




$$L(\boldsymbol{B}) = \{\boldsymbol{Bx} | \boldsymbol{x} \in \mathbb{Z}^n\}$$

$$\boldsymbol{B} = \begin{bmatrix} \underbrace{1}_{b_1} & \underbrace{1}_{b_2} & \underbrace{3}_{b_3} \\ 2 & 2 & 1 \\ 3 & 1 & 2 \end{bmatrix}$$

# Basics of Lattices II

- **Fundamental Parallelepiped**

Let $\boldsymbol{B}$ be an $n \times n$ basis matrix of $L$.

$$P(\boldsymbol{B}) = \{\boldsymbol{Bx} \mid \boldsymbol{x} \in [0,1)^n\}$$

- **Determinant** or **volume** of $L$

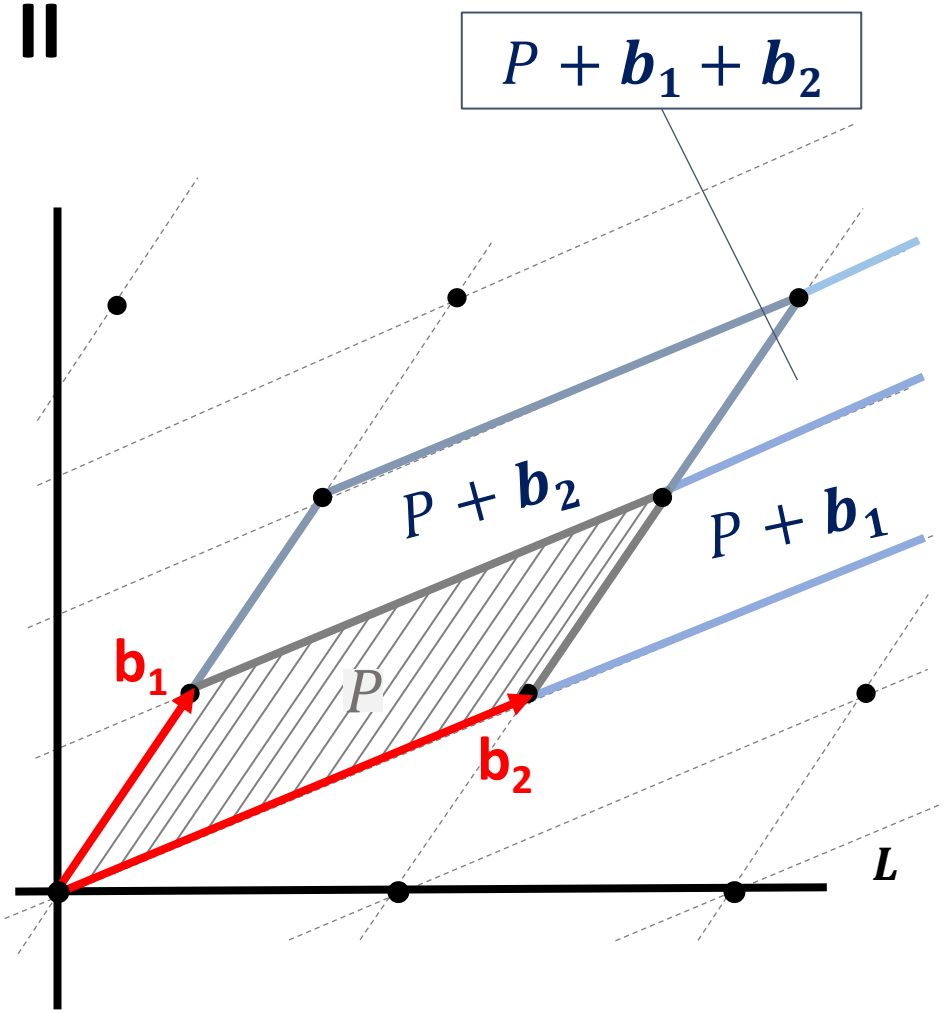
It is the volume of P(**B**), $\det(L) = |\det(\boldsymbol{B})|$

- **Hadamard's Inequality**

Let $\{\boldsymbol{b}_1, \dots, \boldsymbol{b}_n\}$ be a basis of $L$. Then, $\det(L) \leq \|\boldsymbol{b}_1\| \cdots \|\boldsymbol{b}_n\|$, with equality if-f $\boldsymbol{b}_1, \dots, \boldsymbol{b}_n$ are pairwise orthogonal.

G-S

$$\{\boldsymbol{b}_1^*, \dots, \boldsymbol{b}_n^*\} \rightarrow \det(L) = \|\boldsymbol{b}_1^*\| \cdots \|\boldsymbol{b}_n^*\|.$$




$$L(\boldsymbol{B}) = \{\boldsymbol{Bx} | \boldsymbol{x} \in \mathbb{Z}^n\}$$
$$= \{a_1\boldsymbol{b}_1 + a_2\boldsymbol{b}_2 \mid a_1, a_2 \in \mathbb{Z}\}$$

# Basics of Lattices III

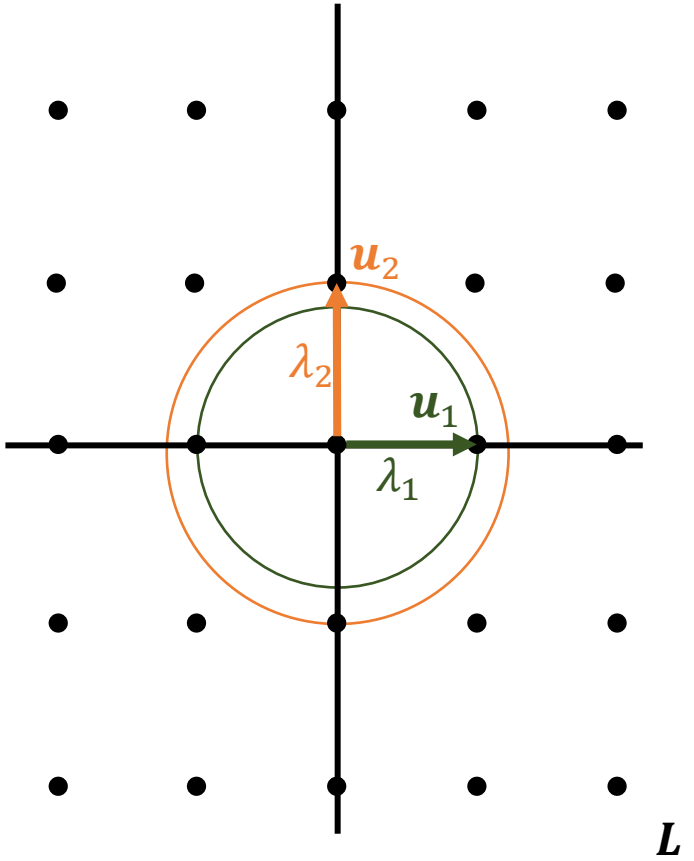
➢ **Successive minima**

For $1 \leq i \leq n$, $\lambda_i$ is the smallest possible positive number s.t. there exist linearly independent vectors $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_i \in L$ satisfying $\|\boldsymbol{u}_i\| \leq \lambda_i$, for $1 \leq j \leq i$.

- $0 < \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$
- $\lambda_1 = \inf_{\boldsymbol{x} \in L/\{\boldsymbol{0}\}} \{\|\boldsymbol{x}\|\}$

  $\downarrow$

  $\lambda_1(L)$



$L \rightarrow$ $n$-dimensional lattice
$\boldsymbol{B} \rightarrow$ some basis matrix of $L$
$\gamma \rightarrow$ Fixed constant $> 1$

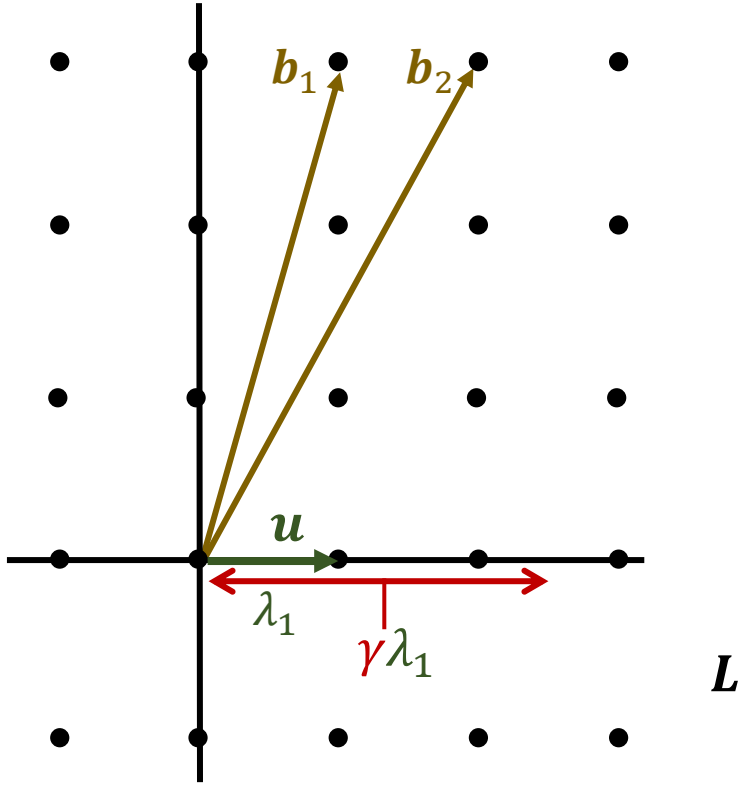
# Hard Lattice Problems I

➢ **Shortest Vector Problem (SVP)**
Given $\boldsymbol{B}$, find nonzero $\boldsymbol{u} \in$ s.t. $\|\boldsymbol{u}\| = \lambda_1(L)$.

➢ **Decisional Shortest Vector Problem (GapSVP)**
Given $\boldsymbol{B}$ and a real number $d > 0$.
Determine whether $\lambda_1(L) \leq \mathrm{d}$ or $\lambda_1(L) > \mathrm{d}$

➢ **Approximate Shortest Vector Problem ($\mathrm{SVP}_\gamma$)**
Given $\boldsymbol{B}$, find nonzero $\boldsymbol{u} \in L$ s.t. $\|\boldsymbol{u}\| \leq \gamma\lambda_1(L)$.

➢ **Approximate Decisional Shortest Vector Problem ($\mathrm{GapSVP}_\gamma$)**
Given $\boldsymbol{B}$ and a real number $d > 0$,
Determine whether $\lambda_1(L) \leq \mathrm{d}$ or $\lambda_1(L) > \gamma\mathrm{d}$



$L \rightarrow n$-dimensional lattice
$\boldsymbol{B} \rightarrow$ some basis matrix of $L$
$\gamma \rightarrow$ Fixed constant $> 1$

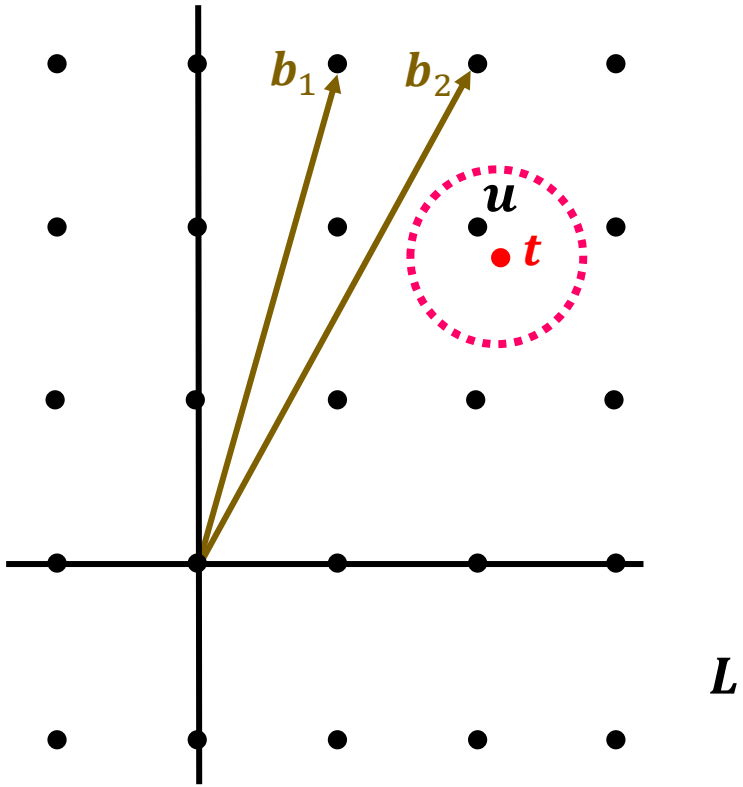
# Hard Lattice Problems II

➢ **Closest Vector Problem (CVP)**
Given $\boldsymbol{B}$ and a point $\boldsymbol{t} \in \mathbb{R}^n$, compute $\boldsymbol{u} \in L$ s.t. $\|\boldsymbol{u} - \boldsymbol{t}\|$ is minimal.

➢ **Decisional Closest Vector Problem (GapCVP)**
Given $\boldsymbol{B}$, a point $\boldsymbol{t} \in \mathbb{R}^n$ and a real number $d > 0$.
Determine whether there exists $\boldsymbol{u} \in L$ s.t. $\|\boldsymbol{u} - \boldsymbol{t}\| \leq \mathrm{d}$ or not.

➢ **Approximate Closest Vector Problem ($\mathrm{CVP}_\gamma$)**
Given $\boldsymbol{B}$ and a point $\boldsymbol{t} \in \mathbb{R}^n$,
compute $\boldsymbol{u} \in L$ s.t. $\|\boldsymbol{u} - \boldsymbol{t}\| \leq \gamma \|\boldsymbol{y} - \boldsymbol{t}\|, \ \forall \boldsymbol{y} \in L.$

➢ **Approximate Decisional Shortest Vector Problem ($\mathrm{GapCVP}_\gamma$)**
Given $\boldsymbol{B}$, a point $\boldsymbol{t} \in \mathbb{R}^n$ and a real number $d > 0$.
Determine whether $\mathrm{dist}(\boldsymbol{t}, L) \leq d$ or $> \gamma d$.

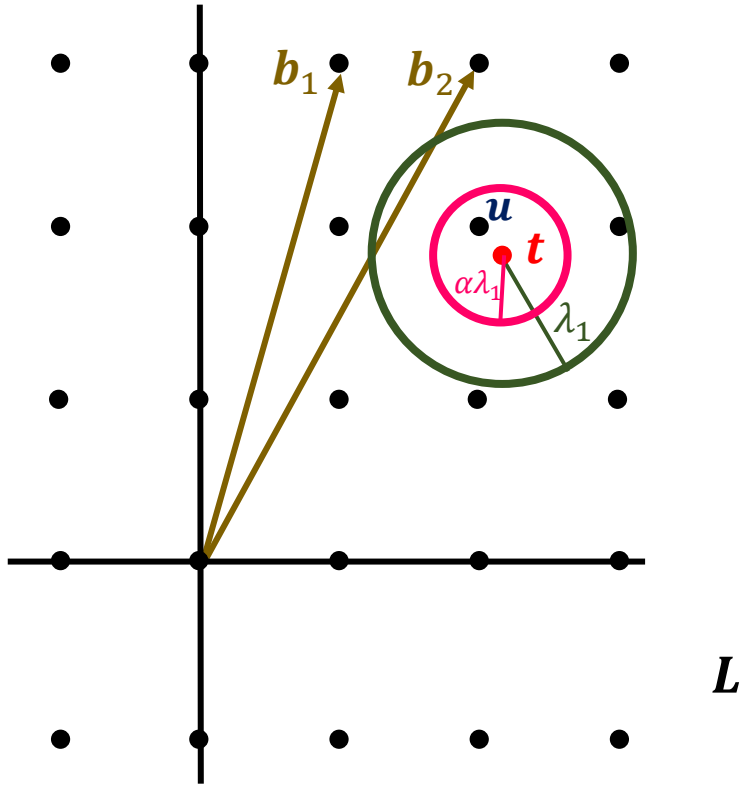


$L \rightarrow$ $n$-dimensional lattice
$\boldsymbol{B} \rightarrow$ some basis matrix of $L$
$\gamma \rightarrow$ Fixed constant $> 1$

# Hard Lattice Problems III

➢ **Approximate Shortest Independent Vectors Problem ($\text{SIVP}_\gamma$)**
Given $\boldsymbol{B}$, output a set $S = \{\boldsymbol{u}_1, \dots, \boldsymbol{u}_n\} \subset L$ of $n$ linearly independent vectors s.t. $\|\boldsymbol{u}_i\| \leq \gamma \lambda_n(L), \forall i$.

➢ **Bounded Distance Decoding Problem ($\text{BDD}_\alpha$)**
Fix $a \in (0,1)$. Given $\boldsymbol{B}$ and a vector $\boldsymbol{t} \in \mathbb{R}^n$ s.t.
$\exists\, u \in L$ with $\|\boldsymbol{u} - \boldsymbol{t}\| \leq \alpha\lambda_1$, compute $\boldsymbol{u}$.



# 1st Approach - Bounds and Heuristics

➢ **Minkowski's Theorem** $\gamma \lesssim \frac{n}{\pi e}$

There is a constant $\gamma$ so that for all lattices $L$ of dimension $n$:

a. ∃ nonzero $\boldsymbol{u} \in L$ s.t. $\|\boldsymbol{u}\| \leq \gamma^{1/2}\det(L)^{1/n}$.

b. ∃ basis $\boldsymbol{u}_1, \dots, \boldsymbol{u}_n$ for $L$ s.t. $\|\boldsymbol{u}_1\| \cdots \|\boldsymbol{u}_n\| \leq \gamma^{n/2}\det(L)$.

} Upper bounds for SVP, SIVP

➢ **Gaussian Heuristic (GH)**

Let $L(B) \subset \mathbb{R}^n$ be a random lattice, with $n$ sufficiently large.

**SVP:** We expect that the smallest nonzero vector in $L$ satisfies

$$\lambda_1(L) = \min_{\boldsymbol{u} \in L/\{\boldsymbol{0}\}} \|\boldsymbol{u}\| \approx \sqrt{\frac{n}{2\pi e}}\det(L)^{1/n}$$

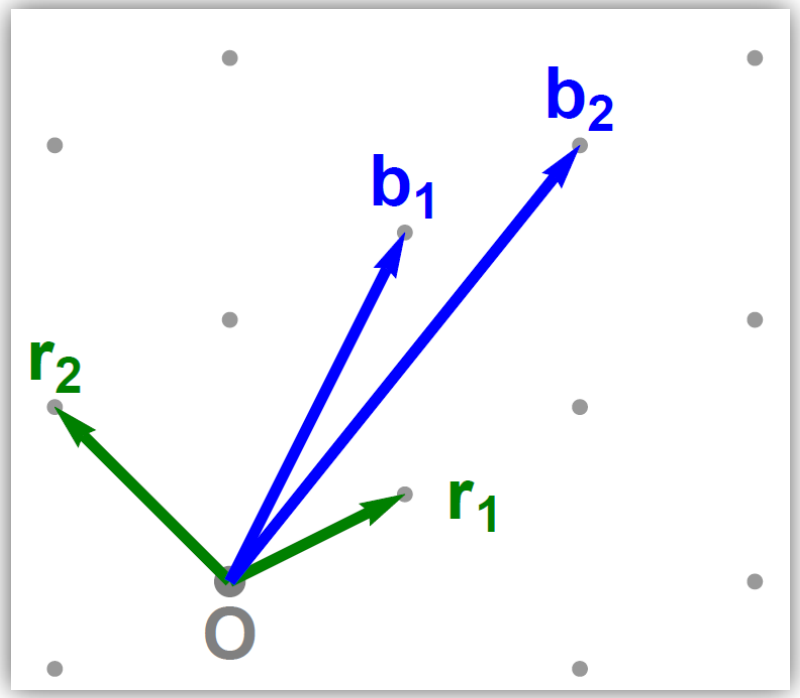
**CVP:** Let $t \in \mathbb{R}^n$ be a random target point. Then, we expect

$$\min_{\boldsymbol{u} \in L} \|\boldsymbol{u} - \boldsymbol{t}\| \approx \sqrt{\frac{n}{2\pi e}}\det(L)^{1/n}$$

# 1st Approach - Good and Bad Bases

➢ **Hadamard's Inequality**

Let $\{\boldsymbol{b}_1, \dots, \boldsymbol{b}_n\}$ be a basis of $L$. Then, $\det(L) \leq \|\boldsymbol{b}_1\| \cdots \|\boldsymbol{b}_n\|$, with equality if-f $\boldsymbol{b}_1, \dots, \boldsymbol{b}_n$ are pairwise orthogonal.




+

Looking for small vectors

⇓

❖ **Lattice Bases**

**"Good" Basis** ← → **"Bad" Basis**

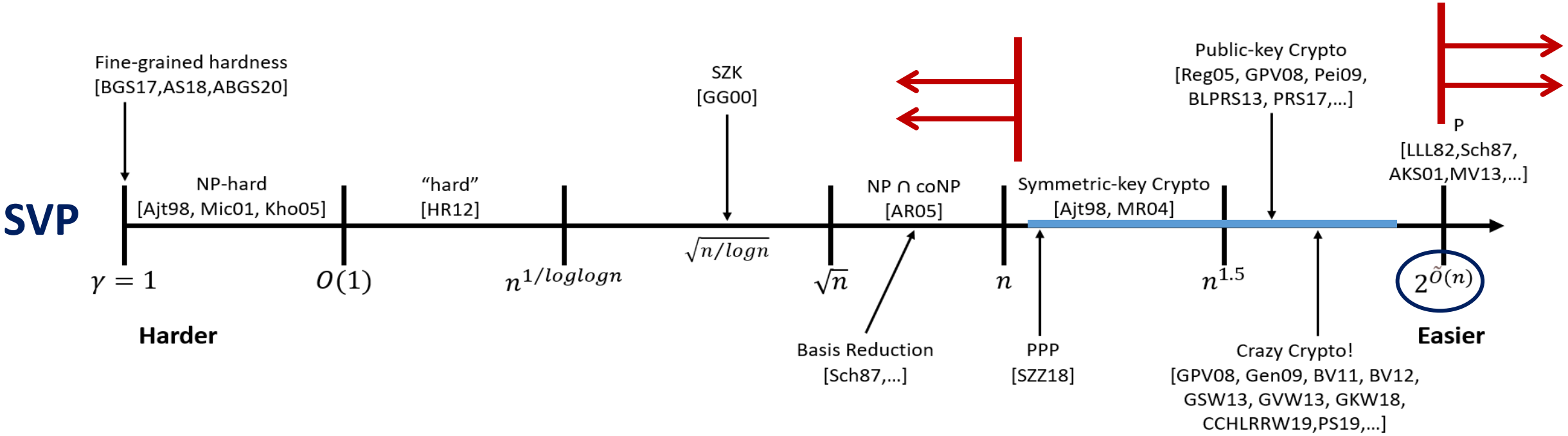
**Lattice Basis Reduction** → Lagrange Algorithm / LLL Algorithm / BKZ Algorithm

LLL Algorithm → $\|\boldsymbol{b}_1\| \leq 2^{(n-1)/2}\lambda_1$

⇓

LLL <u>efficiently</u> solves $SVP_\gamma$, for $\gamma = 2^{O(n)}$

# Complexity of lattice problems



# Complexity of lattice problems

- **Reductions**

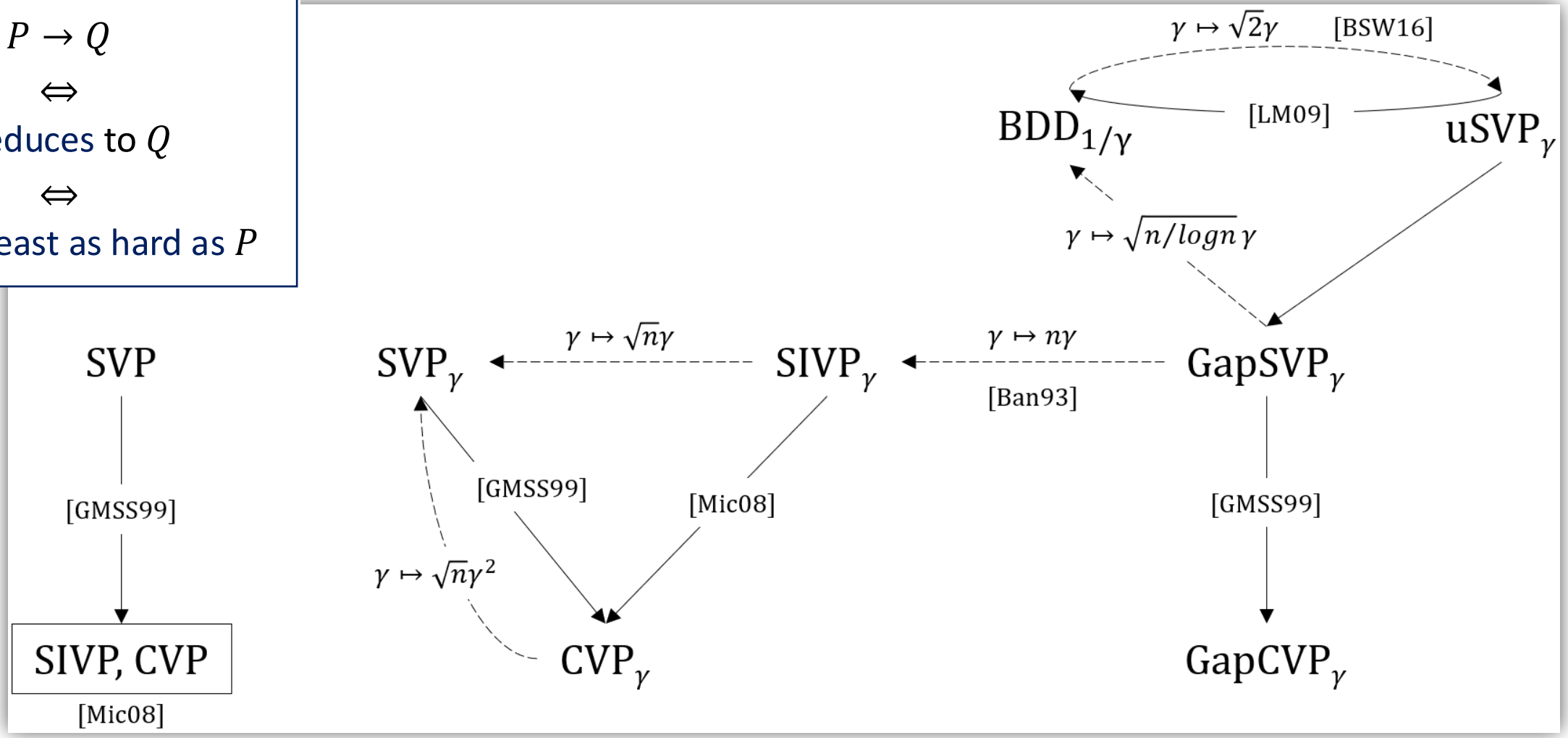
$P \rightarrow Q$

$\Leftrightarrow$

$P$ reduces to $Q$

$\Leftrightarrow$

$Q$ is at least as hard as $P$



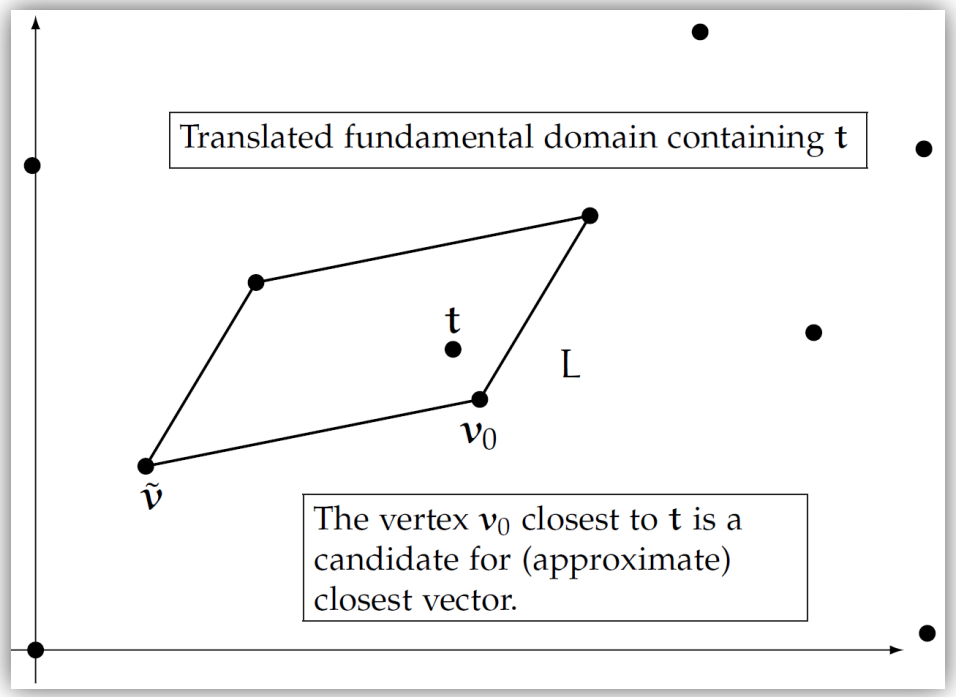
$L \rightarrow n$-dimensional lattice

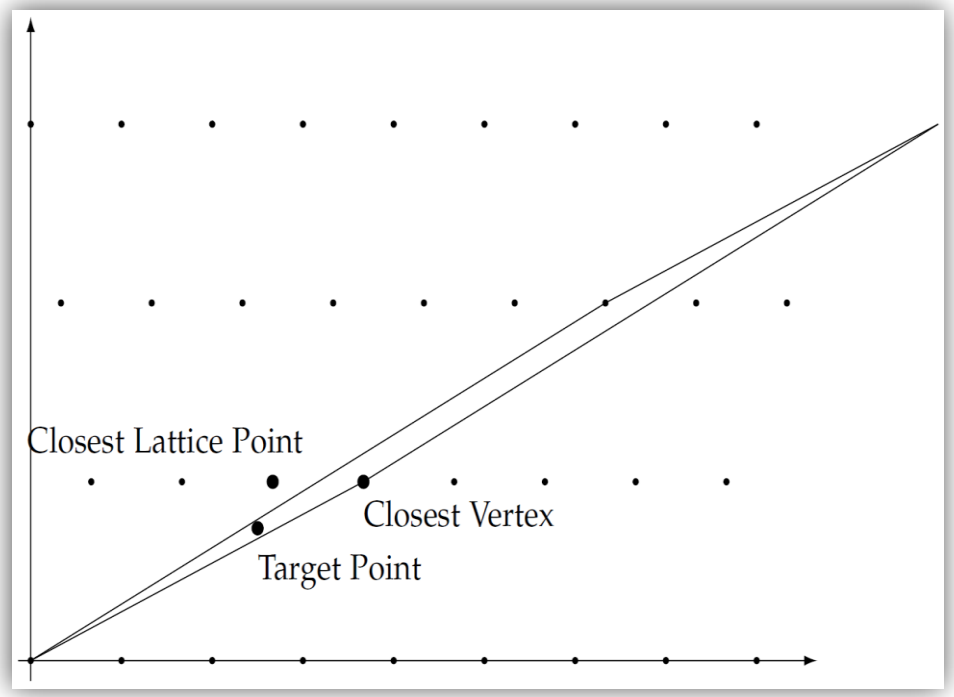
# Solving CVP: Babai's Rounding

➢ **Closest Vector Problem (CVP)**

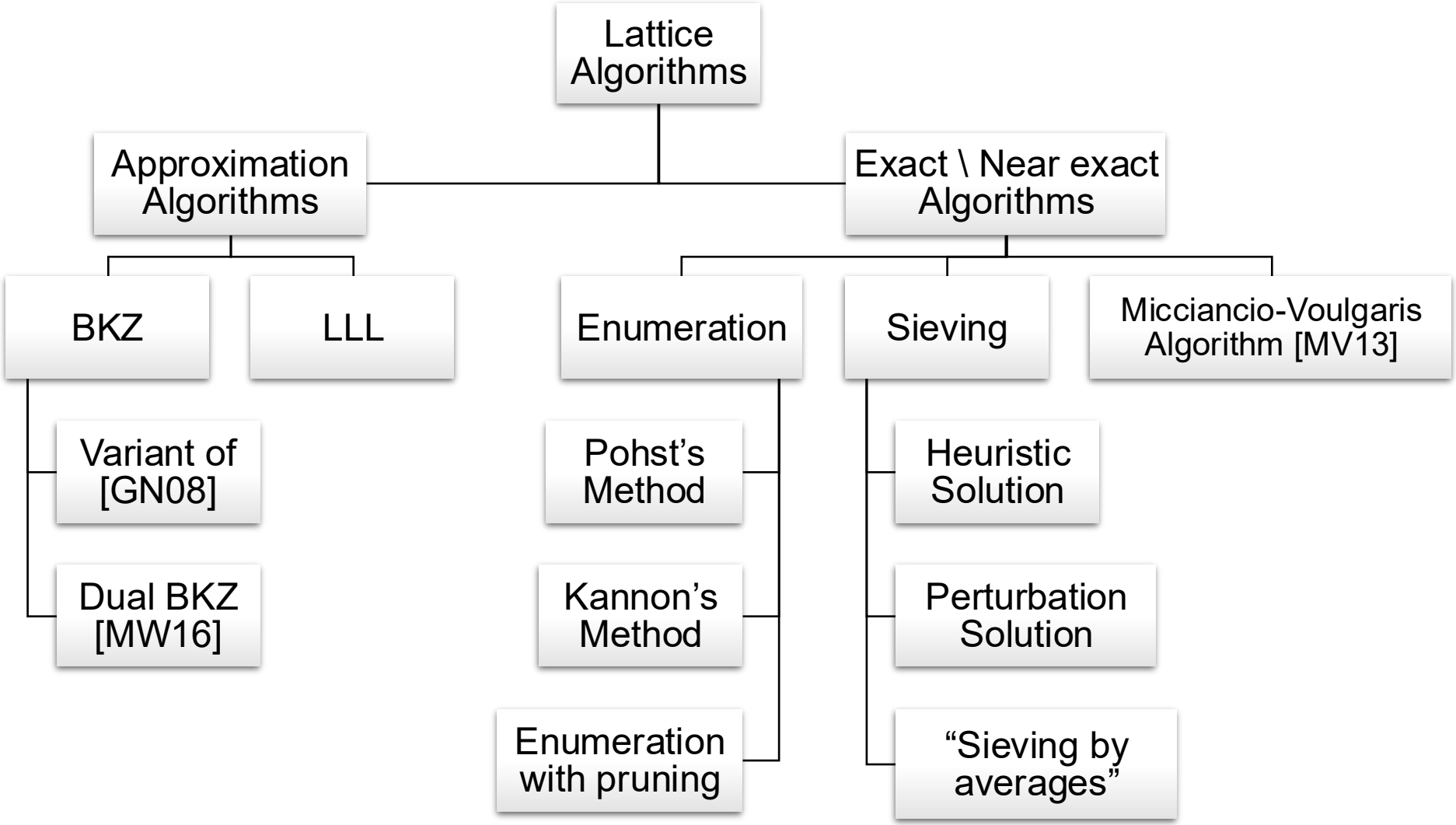
Given a basis $\{\boldsymbol{b}_1, \dots, \boldsymbol{b}_n\}$ and a point $\boldsymbol{t} \in \mathbb{R}^n$, compute $\boldsymbol{u} \in L$ s.t. $\|\boldsymbol{u} - \boldsymbol{t}\|$ is minimal.

$$\boldsymbol{t} = \sum_{i=1}^{n} t_i \boldsymbol{b}_i \longrightarrow \boldsymbol{u} = \sum_{i=1}^{n} \lfloor t_i \rceil \boldsymbol{b}_i$$



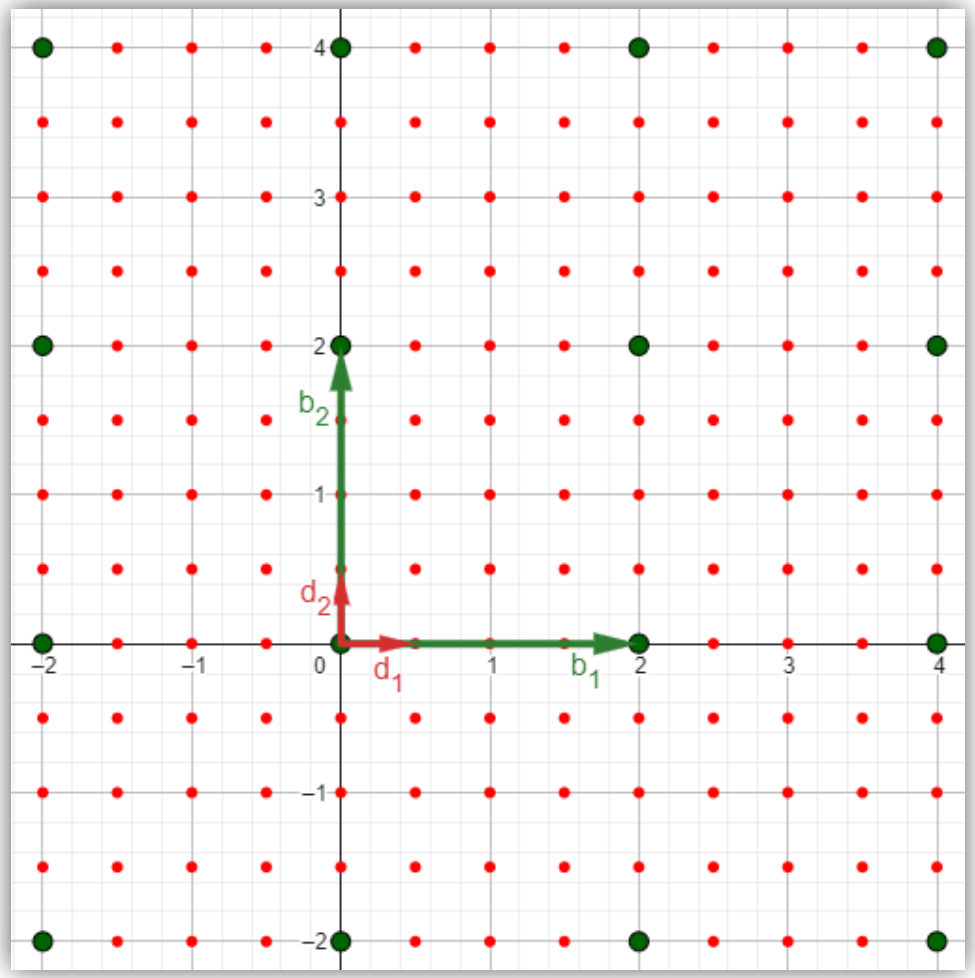



# Solving SVP


Lattice Algorithms
Approximation Algorithms
Exact \ Near exact Algorithms
BKZ
LLL
Enumeration
Sieving
Micciancio-Voulgaris Algorithm [MV13]
Variant of [GN08]
Dual BKZ [MW16]
Pohst's Method
Kannon's Method
Enumeration with pruning
Heuristic Solution
Perturbation Solution
"Sieving by averages"

# Basics of Lattices IV

Dual of $2\mathbb{Z}^2$ is $\frac{1}{2}\mathbb{Z}^2$




## Dual Lattice

Let $L \subseteq \mathbb{R}^m$ be a lattice.

Let $V \subseteq \mathbb{R}^m$ the $\mathbb{R}$-vector space spanned by the vectors in $L$.

$$L^* = \{\boldsymbol{y} \in V \mid \langle \boldsymbol{x}, \boldsymbol{y} \rangle \in \mathbb{Z}, \quad \forall \boldsymbol{x} \in L\}$$

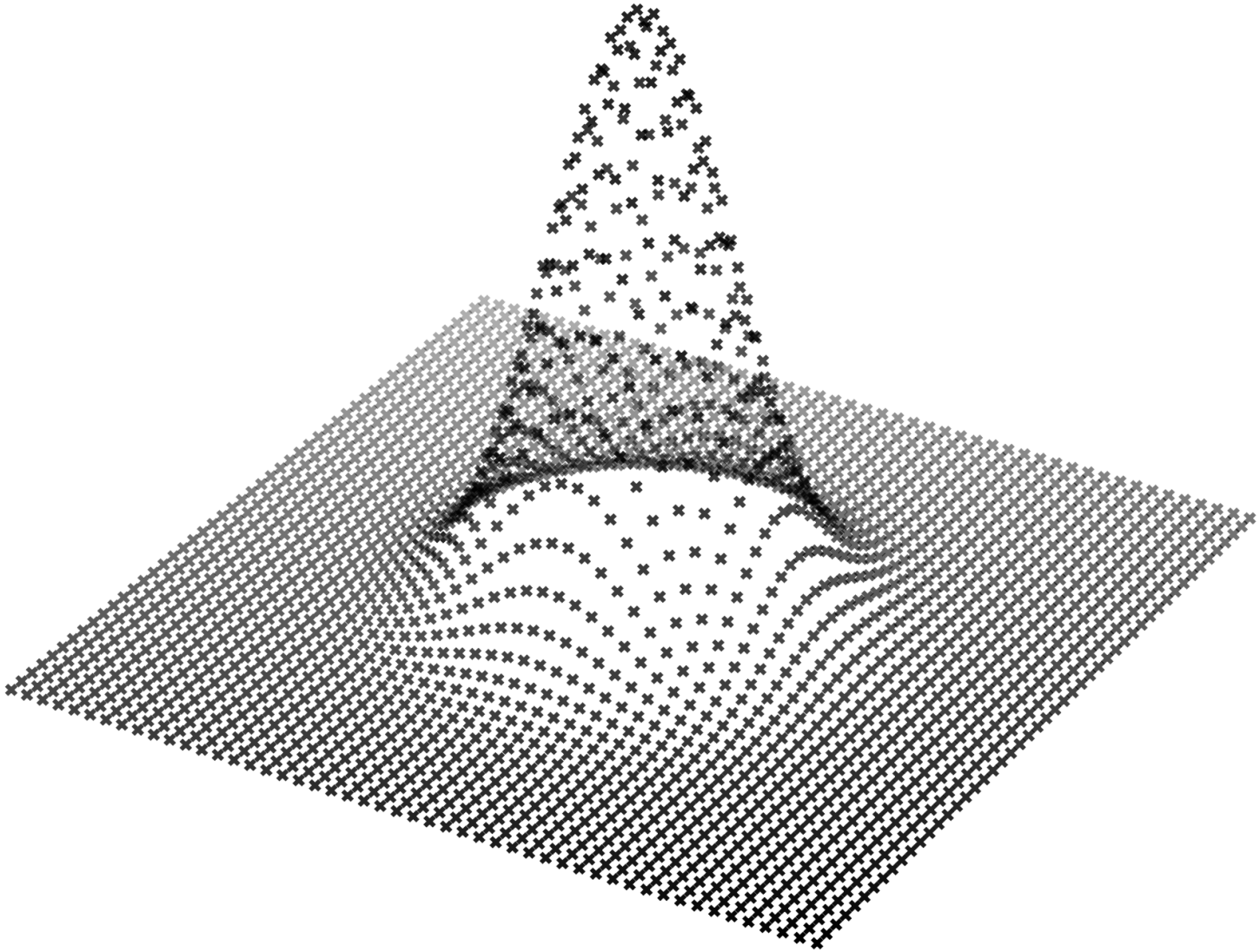
- Dual of $\mathbb{Z}^n$ is $\mathbb{Z}^n$.
- Dual of $cL$ is $\frac{1}{c}L^*$, for $c > 0$ and $L$ lattice.

## Proposition

(i) $L \to$ basis $\boldsymbol{B} \quad \Rightarrow \quad L^* \to$ basis $\boldsymbol{D} = \boldsymbol{B}(\boldsymbol{B}\boldsymbol{B})^{-1}$.

(ii) $\det(L^*) = 1/\det(L)$.

# Learning With Errors



q → positive integer

# Lattices used in Cryptography

- **$q$-ary Lattice**

A lattice $L$ of dimension $m$ satisfying $q\mathbb{Z}^m \subseteq L \subseteq \mathbb{Z}^m$.

- **Parity Check Lattice**

Let $\boldsymbol{A} \in \mathbb{Z}_q^{n\times m}$ be a matrix with coefficients in $\mathbb{Z}_q$.

$$L_q^{\perp}(\boldsymbol{A}) = \{\boldsymbol{x} \in \mathbb{Z}^m \mid \boldsymbol{A}\boldsymbol{x} \equiv 0 \pmod q\}$$

- **Row-generated Lattice**

Let $\boldsymbol{A} \in \mathbb{Z}_q^{n\times m}$ be a matrix with coefficients in $\mathbb{Z}_q$.

$$L_q(\boldsymbol{A}) = \{\boldsymbol{z} \in \mathbb{Z}^m \mid \boldsymbol{z} = \boldsymbol{A}^{\boldsymbol{T}}\ \mathbf{s} \bmod q,\ \text{for } \boldsymbol{s} \in \mathbb{Z}^n\}$$

crypto

↓

Hard Problems over their random instances

↓

Hard Problems over any lattice

Worst-case to average-case reduction

$n, q \rightarrow$ positive integer

$\beta \rightarrow$ positive real

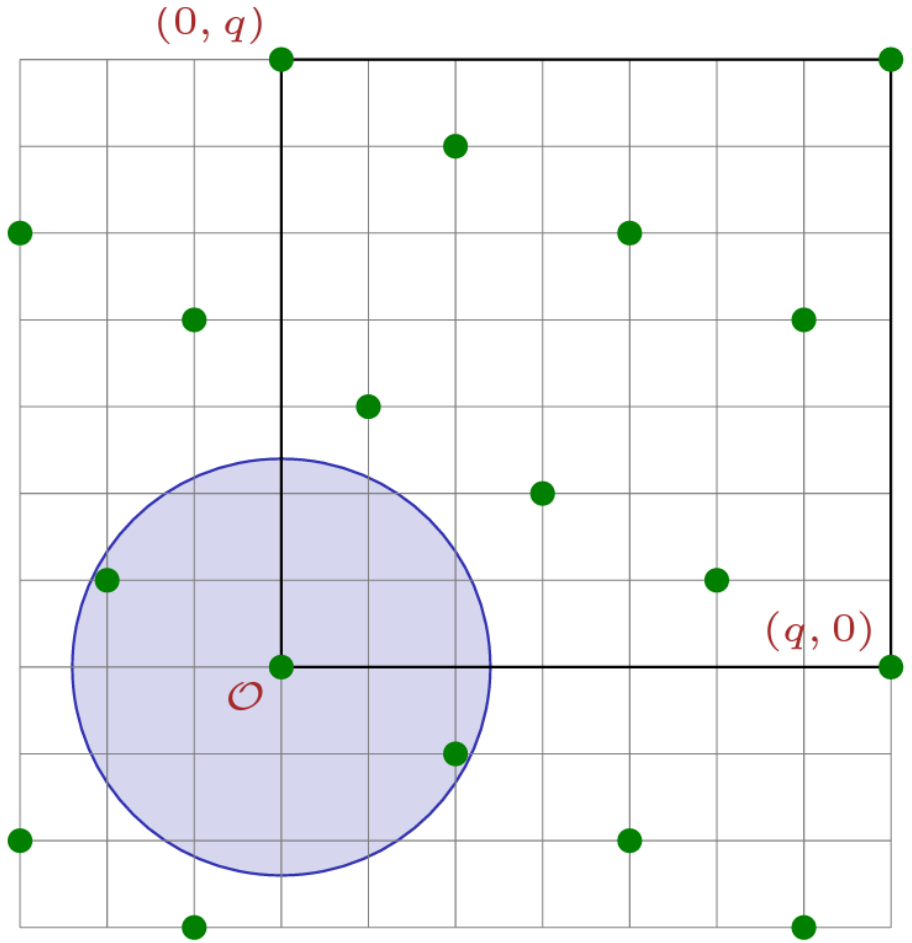
# Shortest Integer Solution (SIS)

$$\mathbf{A} = \begin{pmatrix} | & & | \\ \mathbf{a}_1 & \cdots & \mathbf{a}_m \\ | & & | \end{pmatrix}$$

$\boldsymbol{A} \in \mathbb{Z}_q^{n \times m}$ chosen uniformly at random

➢ **SIS problem**

Find a nonzero vector $z \in \mathbb{Z}^m$ s.t. $\|z\| \leq \beta$ & $\boldsymbol{Az} = 0 \bmod q$.




➢ **Parity Check Lattice**

Let $\boldsymbol{A} \in \mathbb{Z}_q^{n \times m}$ be a matrix with coefficients in $\mathbb{Z}_q$.

$$L_q^{\perp}(\boldsymbol{A}) = \{\boldsymbol{x} \in \mathbb{Z}^m \mid \boldsymbol{Ax} \equiv 0 \pmod q\}$$

↓

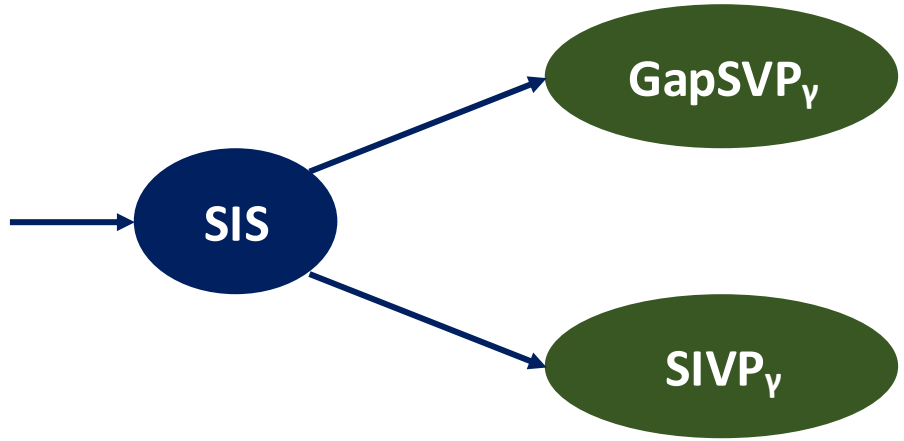
SIS ↔ Average-Case SVP

# Hardness of SIS

**Theorem 3.1.** For any $m = poly(n)$, any $\beta > 0$, and any sufficiently large $q \geq \beta \cdot poly(n)$, solving $\mathrm{SIS}_{n,q,\beta,m}$ with non-negligible probability is at least as hard as solving $\mathrm{GapSVP}_\gamma$ and $\mathrm{SIVP}_\gamma$ (among others) on arbitrary $n$-dimensional lattices (i.e. in the worst case) with overwhelming probability, for some $\gamma = \beta \cdot poly(n)$.

**Crypto**

One-Way Functions,
Collision-Resistant,
Hash Functions,…



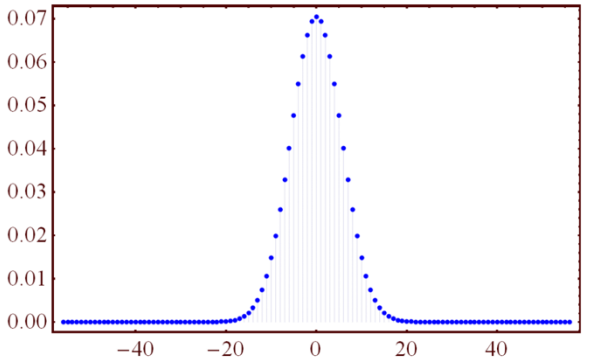
Modulus $q > 2$

"Error rate" $a \ll 1$

# Learning With Errors (LWE)

$$\mathbf{A} = \begin{pmatrix} | & & | \\ \mathbf{a}_1 & \cdots & \mathbf{a}_m \\ | & & | \end{pmatrix}$$

$\boldsymbol{A} \in \mathbb{Z}_q^{n \times m}$ chosen uniformly at random

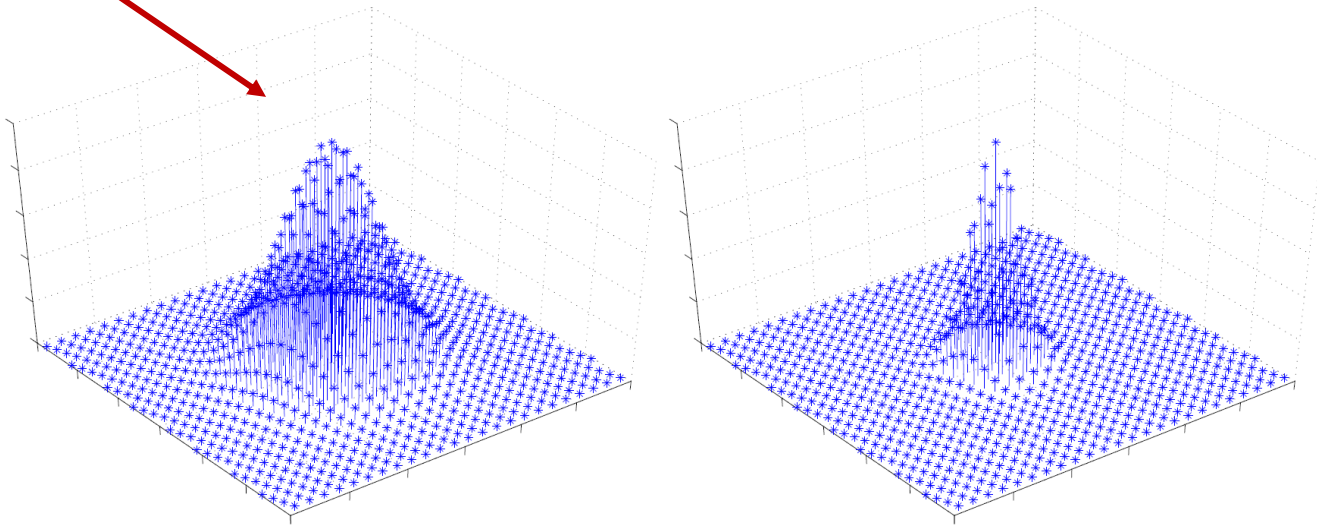



➢ **Search LWE**

Find $\boldsymbol{s} \in \mathbb{Z}_q^n$, given "noisy random inner products" $\rightarrow \boldsymbol{A} \in \mathbb{Z}_q^{n \times m}$, $\boldsymbol{b}^T = \boldsymbol{s}^T \boldsymbol{A} + \boldsymbol{e}^T$, where the errors $e_i$ are chosen from $\chi$ = Discrete Gaussian, param $aq > \sqrt{n}$.

➢ **Decision LWE**

Distinguish $(a_i, b_i)$ from uniform $(a_i, b_i)$ pairs.



➢ **Hermite normal form of LWE**

LWE is not easier if the secret is drawn from the error distribution $\chi$.

Modulus $q > 2$
"Error rate" $a \ll 1$

# Learning With Errors (LWE)

$\boldsymbol{A} \in \mathbb{Z}_q^{n \times m}$ chosen uniformly at random

➢ **Search LWE**

Find $\boldsymbol{s} \in \mathbb{Z}_q^n$, given "noisy random inner products" → $\boldsymbol{A} \in \mathbb{Z}_q^{n \times m}$, $\boldsymbol{b}^T = \boldsymbol{s}^T \boldsymbol{A} + \boldsymbol{e}^T$,
where the errors $e_i$ are chosen from $\chi$ = Discrete Gaussian, param $aq > \sqrt{n}$.

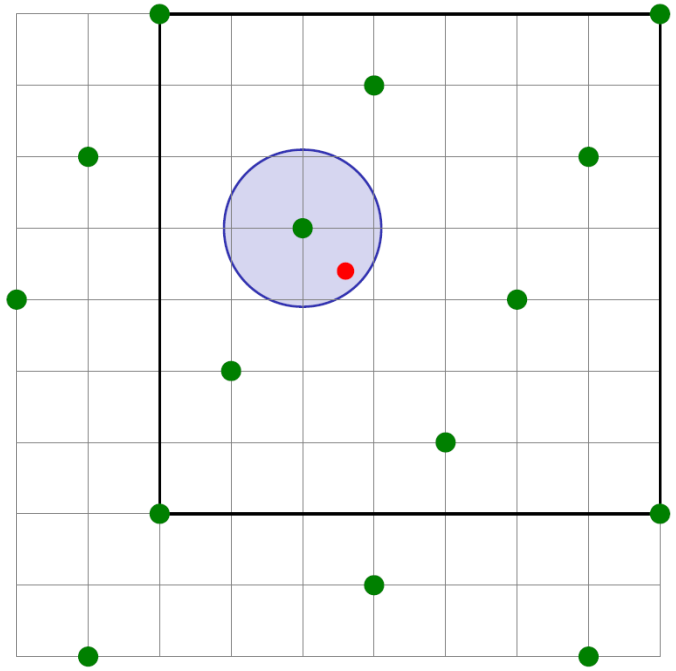
➢ **Row-generated Lattice**

Let $\boldsymbol{A} \in \mathbb{Z}_q^{n \times m}$ be a matrix with coefficients in $\mathbb{Z}_q$.

$L_q(\boldsymbol{A}) = \{\boldsymbol{z} \in \mathbb{Z}^m \mid \boldsymbol{z}^T = \boldsymbol{A}^T \boldsymbol{s} \bmod q, \text{ for } \boldsymbol{s} \in \mathbb{Z}^n\}$

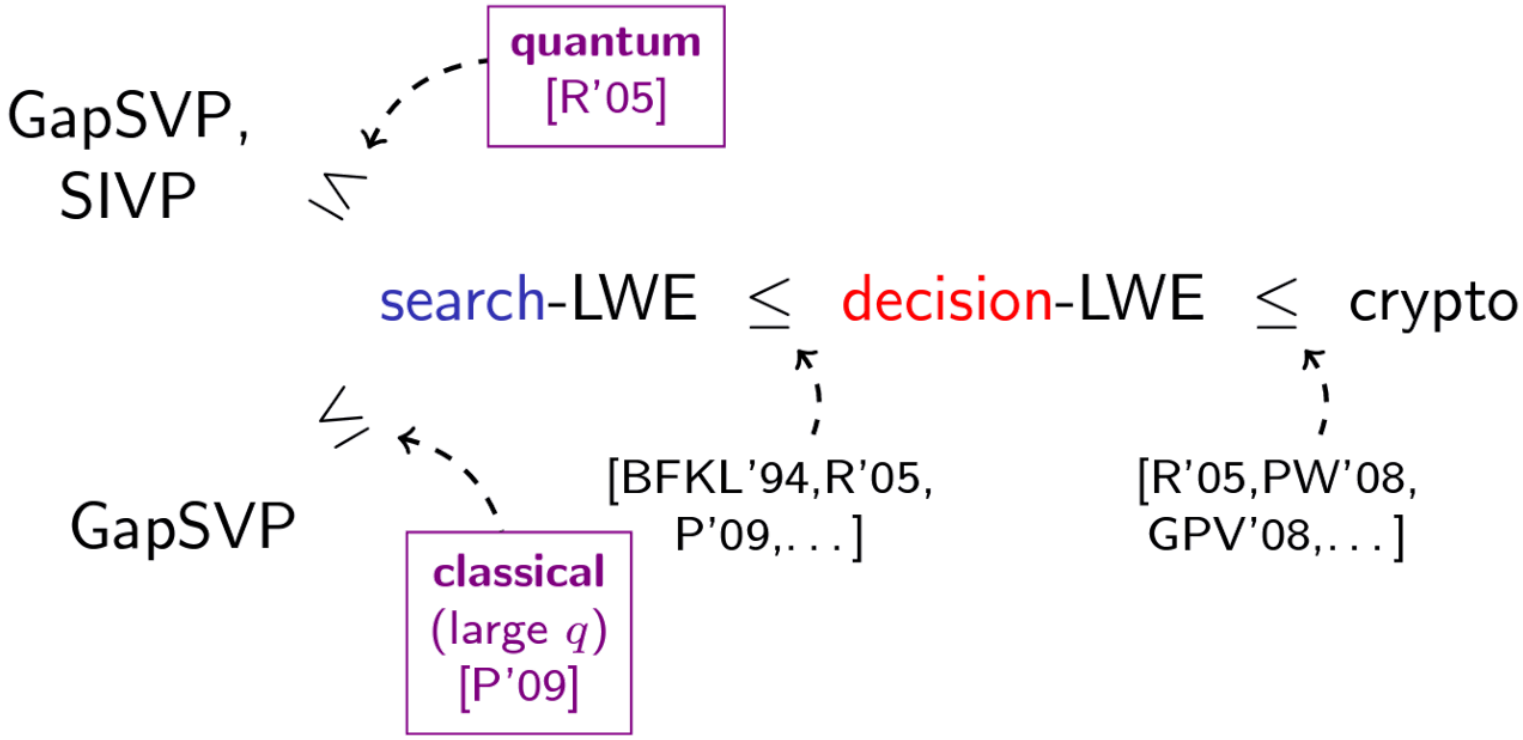
↓

LWE ↔ Average-Case BDD



# Hardness of LWE

**Theorem 3.2.** For any $m = poly(n)$, any modulus $q \leq 2^{poly(n)}$, and any (discretized) Gaussian error distribution $\chi$ of parameter $aq \geq 2\sqrt{n}$ (with $0 < a < 1$), solving the Decision-LWE$_{n,q,\chi,m}$ is at least as hard as quantumly solving GapSVP$_\gamma$ and SIVP$_\gamma$ on arbitrary $n$-dimensional lattices, for some $\gamma = \tilde{O}(n/a)$.



# Regev's LWE Cryptosystem



$s \leftarrow \mathbb{Z}_q^n$

$A \leftarrow \mathbb{Z}_q^{n \times m}$

$\mathbf{x} \leftarrow \{0,1\}^m$



$$b^T = s^T A + e^T$$

public key → $(A, b^T)$

$$\mathbf{u} = A\mathbf{x}$$

Ciphertext "preamble"

$$u' = b^T \mathbf{x} + \text{bit} \cdot q/2$$

"payload"

$$u' - s^T \mathbf{u} \approx \text{bit} \cdot q/2$$



$(A, b^T), (\mathbf{u}, u')$

LWE ↓ $(A, b^T), (\mathbf{u}, u')$

LHL → $m \geq n \log q$

# Compact Cryptosystem



$s \leftarrow \chi^n$

$A \leftarrow \mathbb{Z}_q^{n \times n}$

$r \leftarrow \chi^n$



$$b^T = s^T A + e_1^T$$

public key → $(A, b^T)$

$$u = Ar + e_2$$

Ciphertext "preamble"

$$u' = b^T r + e_3 + \text{bit} \cdot q/2$$

"payload"

$$u' - s^T u \approx \text{bit} \cdot q/2$$



$(A, b, u, u')$

↓ LWE (HNF) x2

$(A, b, u, u')$

Practical limitations of LWE

public data size, speed

↓

Solution

Use structured lattices

# Ring-LWE

- Change $\mathbb{Z}_q^n$ to $R_q := \mathbb{Z}_q[X]/\langle X^n + 1\rangle$, for $n$ a power of 2.
  - Elements of $R_q$ are $\deg < n$ polynomials with $\bmod q$ coefficients.
  - Operations in $R_q$ are very efficient using FFT-like algorithms.

- **Search Ring-LWE**

Find secret ring element $\boldsymbol{s} \in R_q$, given $m$ independent samples

$$a_1 \leftarrow R_q, \quad b_1 = s \cdot a_1 + e_1 \in R_q$$
$$a_2 \leftarrow R_q, \quad b_2 = s \cdot a_2 + e_2 \in R_q$$
$$\vdots \qquad\qquad \vdots$$

where $e_i \leftarrow \chi$, with $\chi =$ Discretized Gaussian in the canonical embedding of $R := \mathbb{Z}[X]/\langle X^n + 1\rangle$.

It is a homomorphism into $\mathbb{C}$, with which we define geometry on the ring.

- **Decision Ring-LWE**

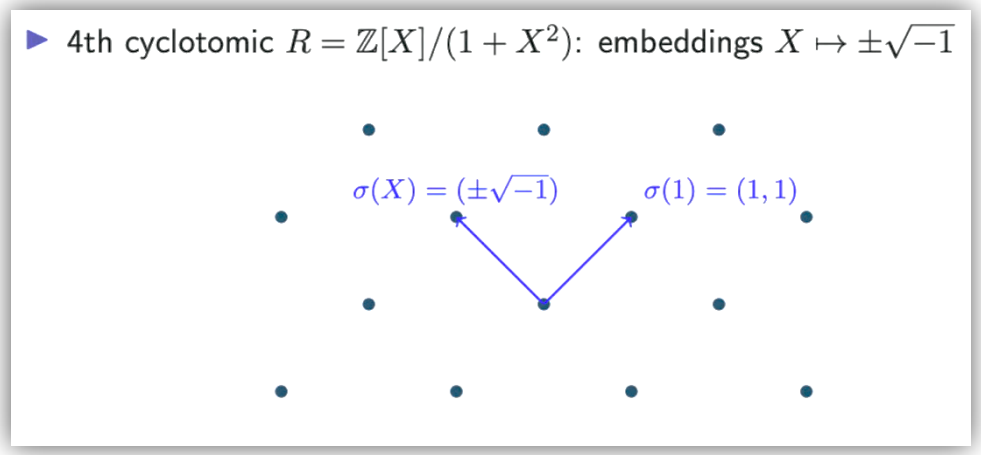
Distinguish $(a_i, b_i)$ from uniform $(a_i, b_i) \in R_q \times R_q$.

# Ring-LWE

➢ **Ideal Lattice**

A lattice corresponding to an ideal $I$ in a ring $R$, under some fixed choice of geometric embedding.

❖ Multiplicative closure of ideals → Extra structure



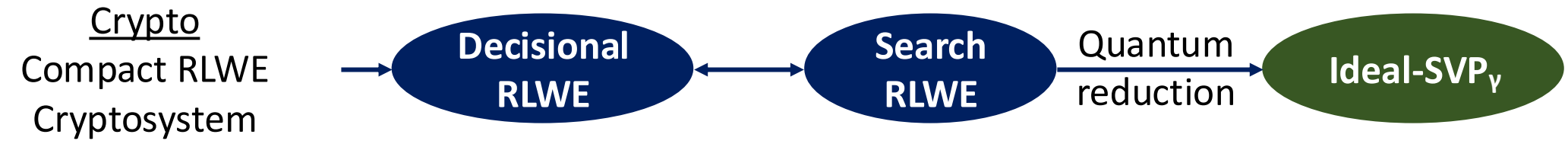

➢ **Problems in Ideal Lattices**

Viewing ideals as lattices allows us to extend lattice problems to ideals giving us the corresponding problems restricted to ideal lattices.

- Ideal-SVP$_\gamma$
- Ideal-SIVP$_\gamma$
- Ideal-GapSVP$_\gamma$

# Hardness of Ring-LWE

**Theorem 4.1.** For any $m = poly(n)$, cyclotomic ring $R$ of degree $n$ (over $\mathbb{Z}$), any appropriate modulus $q$, and error distribution $\chi$ of error rate $a < 1$, solving the Decision-Ring-LWE$_{q,\chi,m}$ problem is at least as hard as quantumly solving Ideal-SVP$_\gamma$ on arbitrary ideal lattices in $R$, for some $\gamma = poly(n)/a$.



# Module-LWE

➢ **Module**

An algebraic structure generalizing rings and vector spaces.

➢ **Module lattices**

- Correspond to finitely generated modules over the ring of integers of a number field
- Generalize both arbitrary lattices and ideal lattices.

➢ **Module-LWE**

- For a secret, it uses a vector $\boldsymbol{s} \in R_q^k$ of ring elements.
- Its samples are of the form $(\boldsymbol{a}, b) \in R_q^k \times R_q$, where either $b = \langle \boldsymbol{s}, \mathrm{a} \rangle + e \bmod q$, for $e \leftarrow \chi$ or $b \in R_q$ is uniformly random.

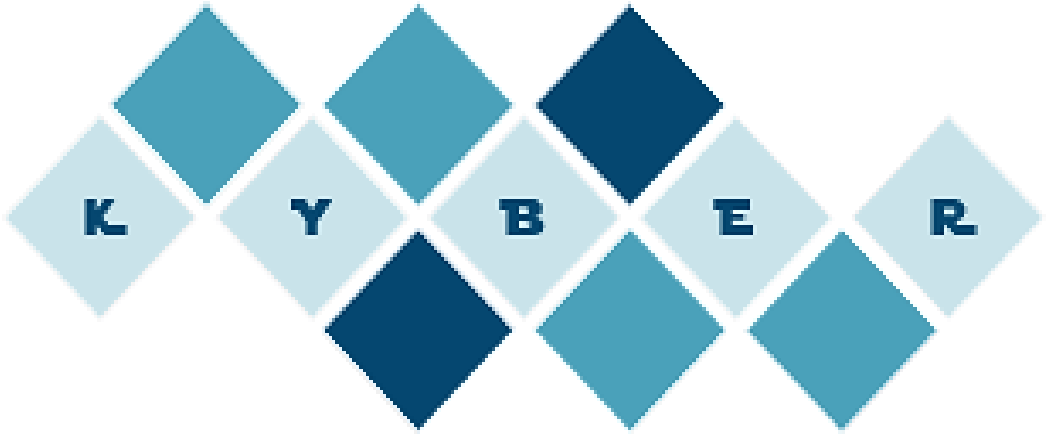
$R = \mathbb{Z}$ → $k$-dimensional LWE problem

$k = 1$ → $RLWE$



# CRYSTALS-Kyber



*FO = Fujisaki-Okamoto

# Creating Kyber

- Compact MLWE Cryptosystem
- NTT in the ring $\mathrm{R_q} = \mathbb{Z}_q[X]/\langle X^n + 1\rangle$
- Compression/Decompression
- $\chi$ = Centered Binomial Distribution

$$\mathrm{Sample}(a_1, \ldots, a_\eta, b_1, \ldots, b_\eta) \leftarrow \{0,1\}^{2\eta} \text{ and output } \sum_{i=1}^{\eta}(a_i - b_i).$$

IND-CPA PKE → IND-CCA2 KEM

Kyber.CPAPKE → Kyber.CCAKEM

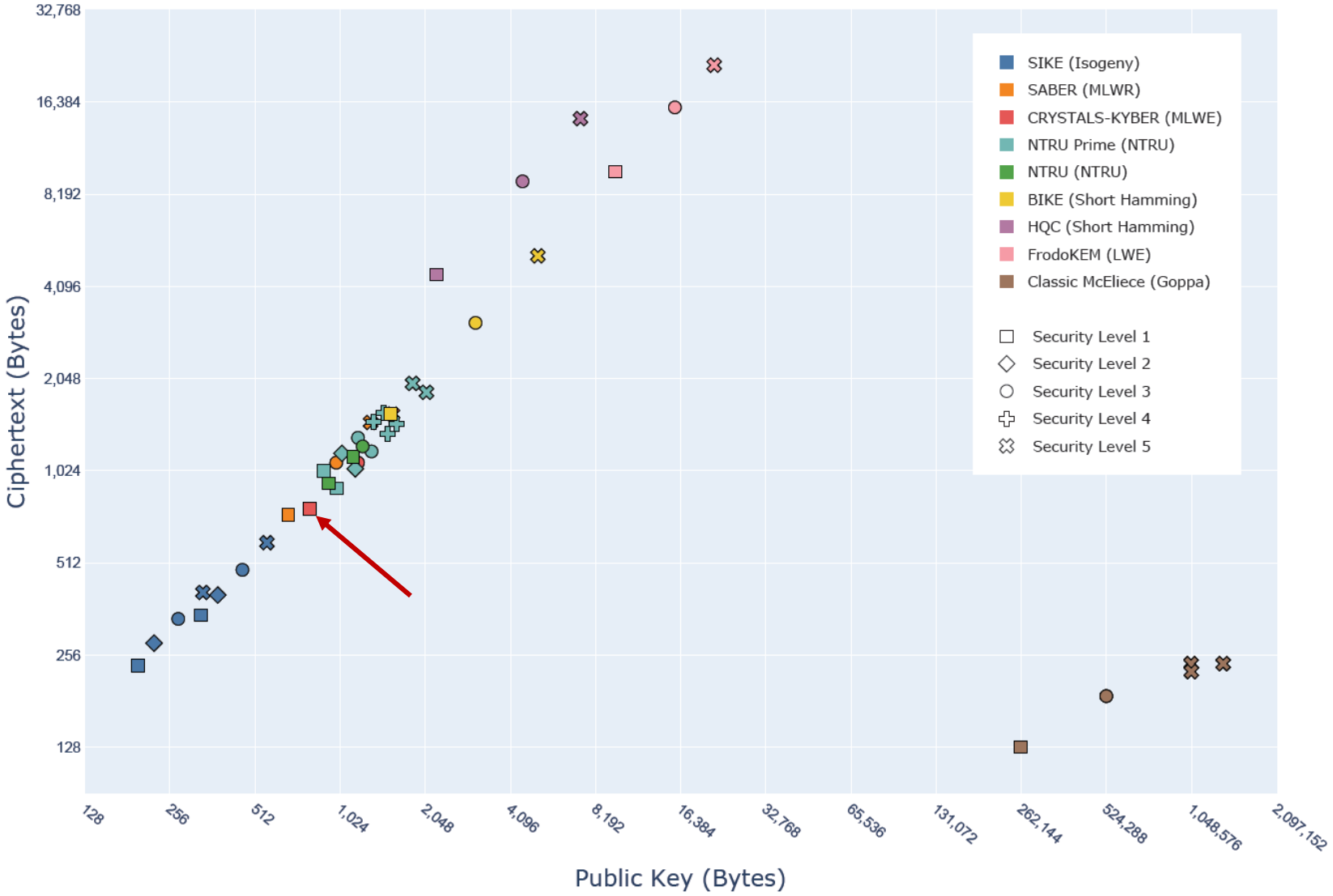
FO Transform

# Kyber's Performance I

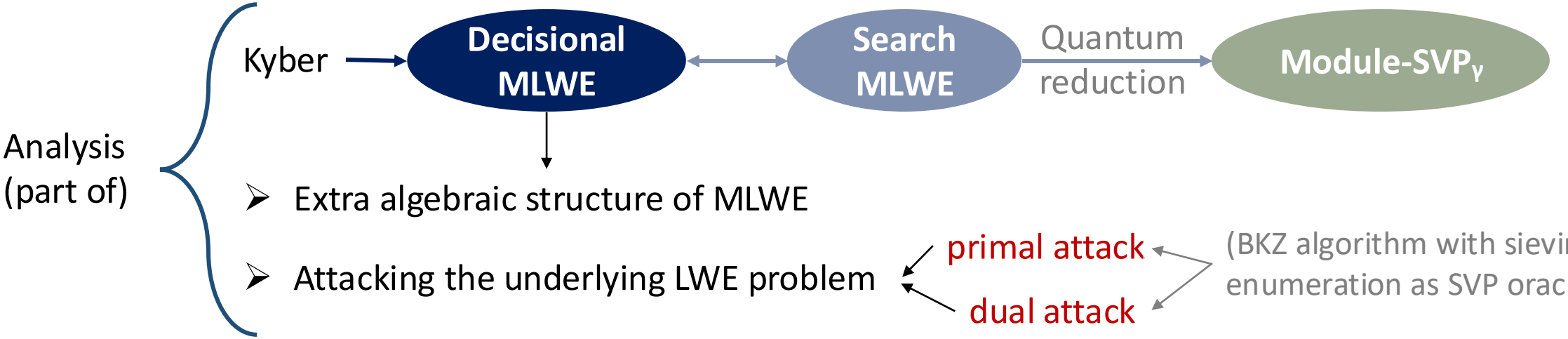
| | | Size keyshares(in bytes) | | Ops/sec (higher is better) | |
|---|---|---|---|---|---|
| Algorithm | PQ | **Client** | **Server** | **Client** | **Server** |
| **Kyber512** | ✅ | 800 | 768 | 50,000 | 100,000 |
| **Kyber768** | ✅ | 1,184 | 1,088 | 31,000 | 70,000 |
| **X25519** | ❌ | 32 | 32 | 17,000 | 17,000 |

- Faster than previous
- Bigger keys than previous

# Kyber's Performance II



# Kyber's Security




| Security Category | Corresponding Attack Type | Example |
|---|---|---|
| 1 | Key search on block cipher with 128-bit key | AES-128 |
| 2 | Collision search on 256-bit hash function | SHA3-256 |
| 3 | Key search on block cipher with 192-bit key | AES-192 |
| 4 | Collision search on 384-bit hash function | SHA3-384 |
| 5 | Key search on block cipher with 256-bit key | AES-256 |

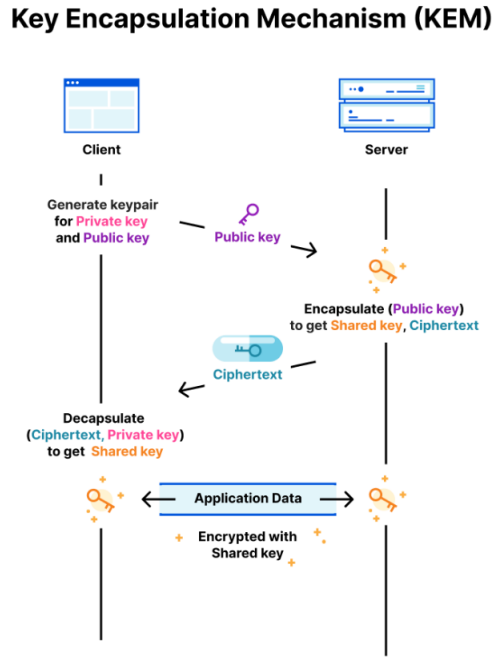
Kyber512 → 1

Kyber768 → 3

Kyber1024 → 5

# Real world application: X25519Kyber768

- X25519Kyber728 is a hybrid post-quantum key exchange protocol designed for use in TLS 1.3.




- Security is achieved if either component is secure.

- Actively used in Google Chrome (from Chrome 116 onwards) and can be enabled in "chrome://flags:

**TLS 1.3 hybridized Kyber support**

This option enables a combination of X25519 and Kyber in TLS 1.3. – Mac, Windows, Linux, ChromeOS, Android, Fuchsia, Lacros

#enable-tls13-kyber

Enabled